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# Optic Nerve Atrophy Conditions Associated With 3D Unsegmented Optical Coherence Tomography Volumes Using Deep Learning

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**IMPORTANCE** Accurate differentiation of optic nerve head (ONH) atrophy is vital for guiding diagnosis and treatment of conditions such as glaucoma, nonarteritic anterior ischemic optic neuropathy (NAION), and optic neuritis. Traditional 2-dimensional assessments may overlook subtle, volumetric changes.

**OBJECTIVE** To determine whether a 3-dimensional (3D) deep learning model trained on unsegmented ONH optical coherence tomography (OCT) scans can reliably distinguish optic atrophy in glaucoma, NAION, optic neuritis, and healthy eyes.

**DESIGN, SETTING, AND PARTICIPANTS** This cross-sectional study used data from multiple clinical trials and referral centers (2008-2025), including randomized trials, longitudinal studies, and referral clinics. Participants included patients with glaucoma, NAION, or optic neuritis and healthy control patients.

**EXPOSURES** Three ResNet-3D-18 models were trained using 5-fold stratified cross-validation. One assessed the full OCT volume, another focused only on the peripapillary region (PPR), and the third considered only the ONH. Identical data splits were used to allow direct performance comparison.

**MAIN OUTCOMES AND MEASURES** Classification accuracy, macro area under the receiver operating characteristic curve (AUC-ROC), precision, recall, and F1 scores, aggregated across all validation folds. Confusion matrices were generated to characterize misclassifications.

**RESULTS** A total of 7014 Cirrus ONH OCT scans from 1382 eyes of glaucoma (n = 113), NAION (n = 311), optic neuritis (n = 163), and healthy controls (n = 715) were analyzed. The mean (SD) age was 54.2 (16.9) years; there were 733 (65%) male patients and 402 (35%) female patients. The entire-volume model achieved 88.9% accuracy (macro AUC-ROC, 0.977; 95% CI, 0.974-0.979) and F1 scores of 0.94, 0.87, 0.78, and 0.91 for glaucoma, NAION, optic neuritis, and healthy eyes, respectively. The PPR-only model reached 85.9% accuracy (AUC-ROC, 0.970; 95% CI, 0.967-0.972), while the ONH-only model attained 87.0% accuracy (AUC-ROC, 0.972; 95% CI, 0.970-0.975). Both achieved F1 scores from 0.71 to 0.94. Optic neuritis presented the greatest classification challenge, misclassified as NAION or healthy when axonal loss was severe or minimal. Activation maps revealed disease-specific regions of interest in the retina, including the retinal nerve fiber layer, ganglion cell layer, and retinal pigment epithelium.

**CONCLUSIONS AND RELEVANCE** Deep learning–based analysis of unsegmented OCT scans reliably distinguished between different forms of optic nerve atrophy, suggesting subtle, disease-specific structural patterns. This automated approach may support diagnostic efforts, guide clinical management of optic neuropathies, and complement less standardized imaging modalities and subjective clinical impressions.

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Supplemental content

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Optic nerve head (ONH) atrophy arises subsequent to various underlying diseases, including glaucoma, nonarteritic anterior ischemic optic neuropathy (NAION), and optic neuritis. Optic atrophy develops because of structural changes in the ONH that are detectable with imaging techniques such as optical coherence tomography (OCT).[1-10] Differentiating among these conditions in clinical settings can be a challenge, particularly in cases where patients present with subclinical disease or have gone untreated or the clinical history is limited. Further, clinicians without specialized neuro-ophthalmology training often struggle to recognize the distinct patterns of atrophy associated with these conditions. Accurate diagnosis is critical to determine the type and extent of ancillary tests needed as well as initiating interventions. Determining if the atrophy is due to an old static event or is part of a potential ongoing disease process is essential to prevent further degeneration or identify associated systemic or neurological diseases. Furthermore, in some cases it may even be possible to reverse the disease course or improve vision dysfunction.

The application of deep learning (DL) techniques to 3-dimensional (3D) OCT volumes presents a shift in how these structural changes can be analyzed. Traditionally, clinical assessments of the ONH often involve the interpretation of individual 2-dimensional B-scans, potentially overlooking subtle, interconnected structural changes that span the entire volume.[11] Additionally, layer segmentation techniques provide valuable quantitative insights, particularly in analyzing thickness of various retinal layers such as retinal nerve fiber layer (RNFL) thickness, macular ganglion cell-inner plexiform layer (GCIPL) thickness, and deeper ONH structures. These methods may still miss more complex, spatially distributed changes that require a more comprehensive volumetric approach.[12] In contrast, DL algorithms are capable of processing and analyzing the entirety of the 3D OCT volume simultaneously, effectively considering every B-scan within the dataset.[13-16] This holistic approach allows the model to extract complex spatial features and relationships that might be imperceptible to the human eye or missed through the examination of isolated cross-sections, offering a more comprehensive and nuanced understanding of the patterns of atrophy present within the ONH. Our previous research successfully demonstrated that this technique can differentiate patterns of ONH swelling using only OCT volumes.[4] Additionally, it revealed that swelling can be distinguished based solely on the peripapillary (PPR) area, rather than being limited to the ONH as previously believed.

The potential clinical impact of this approach is substantial, offering a noninvasive, automated method for disease differentiation that leverages the wealth of information contained within unsegmented 3D OCT volumes. This work not only aims to advance the understanding of disease-specific patterns of ONH atrophy but also to pave the way for improved diagnoses rather than waiting for long-term follow-up evaluations and improved patient management by identifying those who do and do not need intervention.

### Key Points

**Question** Can deep learning reliably distinguish optic nerve head atrophy from glaucoma, nonarteritic anterior ischemic optic neuropathy, and optic neuritis as well as healthy eyes?

**Findings** In this cross-sectional study, a ResNet-3D model analyzing the entire optical coherence tomography (OCT) volume reached 88.9% accuracy, while models analyzing the peripapillary or optic nerve head (ONH) region only attained 85.9% and 87.0% accuracy, respectively (F1 scores, 0.71-0.94), indicating that atrophy signatures reside both within and beyond the ONH.

**Meaning** Automated volumetric deep learning analysis of unsegmented OCT scans may enhance diagnostic accuracy for ONH atrophy, helping clinicians differentiate these conditions more reliably and potentially guiding evaluation and intervention.

## Methods

This study was approved by the institutional review board of the Icahn School of Medicine at Mount Sinai. It required no additional consent because the data used were deidentified and derived from participants who had provided consent for use of their data at multiple study institutions and clinical trials. The study was conducted according to the tenets of the Declaration of Helsinki and is in accordance with Health Insurance Portability and Accountability Act regulations. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline. This study compares the 3 clinical disorders using only the OCT data as the extent of vision loss among the 3 varied widely. The range of visual acuity and visual field data for each group can be found in eTable 1 in Supplement 1.

### OCT Scans

We collected Cirrus SD-OCT 200 × 1024 × 200 ONH volume scans (Zeiss-Meditec) of eyes with glaucoma, NAION, optic neuritis, and healthy eyes. We used both automatically and manually segmented RNFL thicknesses for each group to gain insights into findings.[12] See the eMethods in Supplement 1 for specific analyses undertaken for glaucoma,[17] NAION,[18] optic neuritis, and healthy eyes. Briefly, we included scans from the Variability in Perimetry Study (VIP, NCT01051739), a Quark Pharmaceuticals study (Quark207 trial, NCT02341560), the Mount Sinai Neuro-Ophthalmology clinic, students at the University of Iowa, and healthy eyes from numerous locations.

### Model Architecture

We took a similar approach to model design as in our prior work.[4] We designed 3 models with similar architectures to analyze ONH OCT scans. We created the base architecture in PyTorch and fine-tuned it from pretrained ResNet-3D-18 weights (a variant of the ResNet-18 convolutional neural network architecture adapted for 3D imaging).[19]

We trained our model using the Adam optimizer with a 0.0001 learning rate, which remained constant except for

Table. Summary of Demographic and Clinical Characteristics

| Characteristic | No. of scans | No. of eyes | Age, mean (SD), y | Male patients, No. (%) | Female patients, No. (%) | RNFL thickness, mean (SD), µm |
|---|---|---|---|---|---|---|
| Glaucoma | 1839 | 113 | 61.2 (10.5) | 22 (38) | 36 (62) | 58.6 (12.2) |
| NAION (QRK) | 830 | 311 | 61.0 (7.7) | 234 (74) | 83 (26) | 62.3 (14.3) |
| NAION (clinic) | 522 | 80 | 66.1 (9.6) | 41 (67) | 20 (33) | 63.8 (11.0) |
| Optic neuritis (clinic) | 990 | 163 | 47.3 (13.1) | 43 (27) | 119 (73) | 71.7 (14.1) |
| Healthy (Iowa) | 913 | 254 | 20.1 (1.4) | 127 (100) | 0 | 100.6 (8.4) |
| Healthy (QRK) | 1178 | 352 | 61.2 (7.4) | 245 (70) | 107 (30) | 93.5 (18.7) |
| Healthy (VIP) | 742 | 109 | 61.2 (8.9) | 21 (36) | 37 (64) | NA[a] |

Abbreviations: Iowa, University of Iowa study; NA, not available; NAION, nonarteritic anterior ischemic optic neuropathy; QRK, Quark Pharmaceuticals Quark207 trial; RNFL, retinal nerve fiber layer thickness; VIP, Variability in Perimetry study.

[a] Retinal thickness measurements were not available for healthy eyes from the VIP.

scheduled reductions via a StepLR scheduler, decreasing it 10× every 10 epochs. Default parameters were used throughout training. Our batch size was 4 for both training and validation because of computational constraints when processing OCT data. Additionally, we implemented early stopping with a patience of 5 epochs and a minimum loss improvement threshold of 0.001 to prevent overfitting. We also incorporated gradient accumulation with 4 accumulation steps to stabilize training with a small batch size

We downscaled all OCT volumes to 100 × 256 × 100 to reduce computational demands. We applied data augmentation by normalizing the OCT values, performed random cropping with up to 25% removal from each dimension and repadding the images with black voxels, and applied gaussian smoothing with a $\sigma = 0.2$. The validation data were only normalized and did not undergo any cropping, padding, or smoothing. The final layer outputs the probability of the OCT volume belonging to any 1 of the groups. For all models, we used a 5-fold stratified cross-validation strategy with grouping constraints. This ensured that each fold had similar class distributions (stratification) while keeping all scans from the same eye within the same fold (grouping). For each iteration, we trained on 4-folds and validated on the remaining fold, repeating this process for all 5-folds. This also enabled us to evaluate the entire dataset. Since the same data splits were used for all models, we were able to directly compare their performance on each specific data split.

We created 3 models to evaluate DL discrimination based on different areas of the OCT. We trained model A on the full ONH OCT scan. Model B consisted of the same scans but with a cylindrical mask centered on the ONH at the center of the volume such that only the PPR is visible. To ensure coverage of the ONH, the cylinder was designed with a radius of 25 slices (out of a total width and length of 100 slices) and extended across the entire vertical dimension of the volume. For model C, instead of applying a cylindrical mask in the center of the scan, we masked all regions outside the cylinder, effectively removing information about the PPR (eFigure 1 in Supplement 1).

We extracted activation maps from all models from the last convolutional layer. These maps were averaged across channels to generate 1 activation map per B-scan, resized to match the input dimensions, and overlaid for visual analysis. We used a bespoke interactive visualization tool to explore activation patterns across different slices of each scan.

To evaluate model performance, we computed area under the receiver operating characteristic curve (AUC-ROC), precision, recall, and F1 scores for each class and generated confusion matrices aggregated across all folds.

## Results

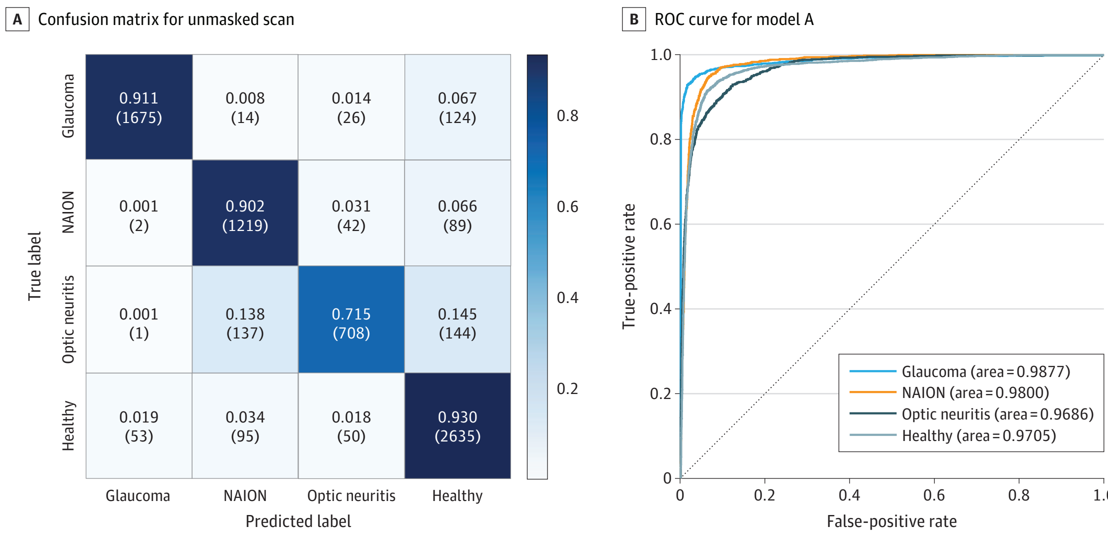
We performed DL on 7014 scans from 1382 eyes with demographics shown in the **Table**. The mean (SD) age was 54.2 (16.9) years; there were 733 (65%) male patients and 402 (35%) female patients.

Over all folds, model A achieved an overall accuracy of 88.9% (range, 86.9%-93.9%). The confusion matrix found F1 scores of 0.94, 0.87, 0.78, and 0.91 for glaucoma, NAION, optic neuritis, and healthy eyes, respectively. We had an overall macro-average AUC-ROC of 0.977 (95% CI, 0.974-0.979) (**Figure 1**).

We found that activation map highlights varied across groups. In glaucoma, they were predominantly focused on the nerve fiber layer just outside the ONH, as well as near the lamina cribrosa. NAION showed activations centered on both the central and more peripheral regions of the retinal pigment epithelium as well as Bruch membrane. Optic neuritis exhibited a combination of peripheral retinal pigment epithelium and RNFL involvement. In contrast, healthy eyes demonstrated activations in the most anterior aspect of the optic nerve as well as both central and peripheral RNFL regions (**Figure 2**).

Model B, which focused exclusively on the PPR, yielded an overall accuracy of 85.9% (range, 83.8%-89.7%) with an F1 score of 0.90, 0.83, 0.72, and 0.89 for glaucoma, NAION, optic neuritis, and healthy eyes, respectively; the macro-average AUC-ROC was 0.970 (95% CI, 0.967-0.972) (**Figure 3**). Model C, which concentrated entirely on the ONH without incorporating surrounding information, reached an overall accuracy of 87.0% (range, 85.7%-88.9%), as well as an F1 score of 0.94, 0.85, 0.71, and 0.89 for glaucoma, NAION, optic neuritis, and healthy eyes, respectively (**Figure 4**). It had a macro-average AUC-ROC of 0.972 (95% CI, 0.970-0.975) (eTable 2 in Supplement 1).

**Figure 1. Confusion Matrix and Receiver Operating Characteristic (ROC) Curve for Model A**




A, Confusion matrix over all folds for the entire unmasked scan. Model A achieved an overall accuracy of 88.9% with a precision of 0.97, 0.83, 0.86, and 0.88, as well as a recall of 0.91, 0.90, 0.72, and 0.93 and F1 scores of 0.94, 0.87, 0.78, and 0.91 for eyes with glaucoma, nonarteritic anterior ischemic optic neuropathy, optic neuritis, and healthy eyes, respectively. B, The ROC curve for model A demonstrates high discriminatory capability across all classes. The macro-average area under the curve is 0.977.

Among the misclassifications of healthy eyes, specific patterns were observed based on the source of the data. Of the 53 healthy scans misclassified as glaucoma, the majority (51) were from the VIP control group and 2 from the University of Iowa group. The 95 healthy eyes misclassified as NAION were entirely from the Quark207 fellow eye group. These misclassified eyes had a lower mean RNFL thickness (72 µm) compared with correctly classified healthy eyes (98 µm), with a difference of −26.6 µm (95% CI, −33 to −21).

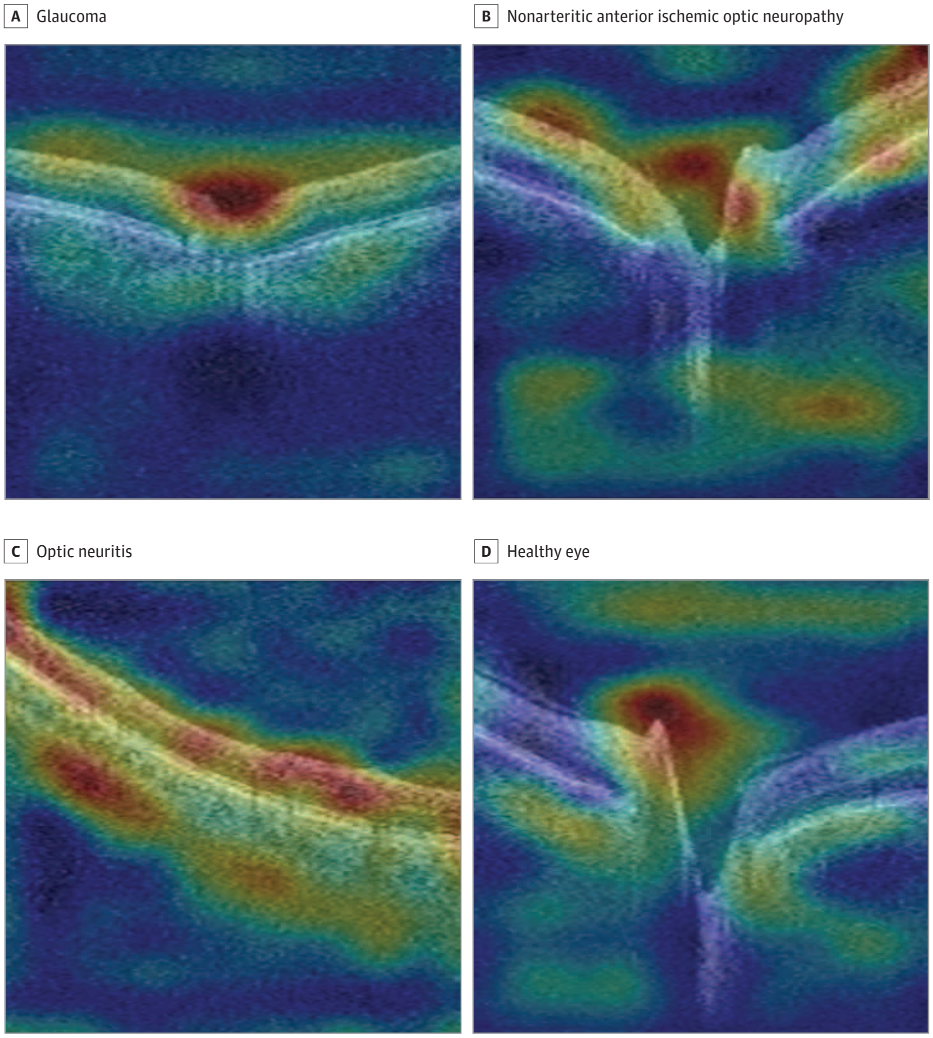
There were 50 healthy eyes misclassified as optic neuritis, with 46 coming from the Quark207 fellow eyes and 4 from the University of Iowa. These had a mean RNFL thickness of 85 µm compared with 98 µm in healthy eyes, with a difference of −13 µm (95% CI, −17 to −10). The 89 NAION eyes misclassified as healthy had a mean RNFL thickness of 78.6 µm vs 62.4 µm in correctly classified NAION eyes, with a difference of 16.2 µm (95% CI, 13.8 to 18.5). NAION eyes misclassified as optic neuritis (n = 42) had a mean RNFL thickness of 63.2 µm compared with 62.4 µm in NAION eyes, with a difference of 0.8 µm (95% CI, −3.5 to 5.0). The 144 optic neuritis eyes misclassified as healthy had a thicker RNFL, averaging 86.7 µm vs 72.6 µm in correctly classified optic neuritis eyes, with a difference of 14.1 µm (95% CI, 12.6 to 15.7), along with a macular GCIPL thickness of 77.9 (6.4) µm and a 10-2 visual field mean deviation of −1.4 ± 1.6 dB. In contrast, optic neuritis eyes misclassified as NAION showed a thinner RNFL, averaging 61.2 µm compared with 72.6 µm, with a difference of −11.4 µm (95% CI, −13.8 to −8.9), as well as a macular GCIPL thickness of 52.7 (15.9) µm and a 10-2 visual field mean deviation of −9.7 (9.7) dB.

## Discussion

Our study evaluated 3 machine learning models, each targeting different relevant regions of the optic nerve, and model A, which used the entire OCT scan, demonstrated the highest overall performance in accuracy, F1 scores, and AUC-ROC across all classes. Notably, models B (PPR alone) and C (ONH alone), which excluded specific areas of the scan, still showed reasonable or very good overall accuracy and discriminative capability. This suggests that atrophic patterns may be differentiable between conditions but also could remain distinctive across various regions of the scan, including non-optic disc areas (PPR) not classically considered using ophthalmoscopy or fundus photos to evaluate these diseases. However, further validation in larger and more demographically diverse datasets is needed to confirm the generalizability of these findings. The overlap in RNFL thickness across disease groups suggests that the model is not solely relying on disparities in thickness measurements to differentiate these conditions but may be learning spatial patterns of retinal atrophy that are associated with these disorders.

Across the 3 models, the classification of optic neuritis presented the greatest challenge, with F1 scores ranging from 0.71 to 0.78. Models A and B misclassified optic neuritis cases roughly equally between NAION and healthy eyes. Grouping misclassifications by the average RNFL thickness suggests that eyes with optic neuritis misclassified as NAION had a thinner RNFL and exhibit a pattern resembling NAION, whereas those with thicker average RNFL appeared similar to healthy eyes. The latter group exhibited minimal axonal thinning or loss, with

Figure 2. Activation Maps Across Different Optic Nerve Conditions




Representative B-scans from 3-dimensional optic nerve head optical coherence tomography scans, with final layer activation maps projected onto 2-dimensional slices to highlight key regions used by the model for differentiation. Each volume consists of 100 B-scan slices, with select slices shown for eyes with (A) glaucoma (slice 37), (B) nonarteritic anterior ischemic optic neuropathy (slice 45), (C) optic neuritis (slice 98), and (D) healthy eyes (slice 50), each of which exhibit distinct activation patterns.

a very mildly depressed central visual field on average. Regardless, since the majority of optic neuritis eyes are correctly identified, it is likely that most eyes with NAION and optic neuritis exhibit predominantly distinct spatial patterns of atrophy after an episode.

Model C provided further insight into the key areas critical for differentiation. While model B nearly evenly misclassified optic neuritis eyes between NAION and healthy, model C misclassified substantially more optic neuritis eyes as healthy rather than NAION. We also found most optic neuritis activations in the PPR. This suggests that the PPR is a key region of optic neuritis–related retinal atrophy and is more critical for DL differentiation than findings in the ONH region.

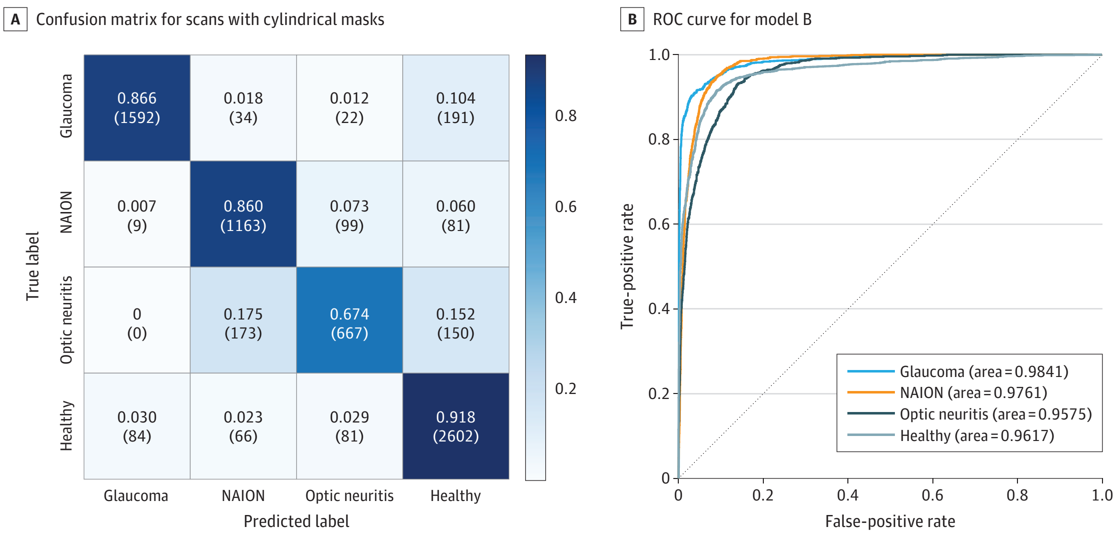
The models misclassified glaucoma at the same rate in both model A and model C. However, since glaucoma is predominantly thought to affect the ONH, it is notable that the PPR is still distinct from other diseases even with a slightly lower classification rate.

In general, across all the models, the most common misclassification was categorizing a diseased eye as healthy. This may reflect eyes that experienced a milder clinical course, resulting in little to less axonal loss or thinning or secondary retinal changes. Alternatively, at some of the earlier time points from acute damage, there may not have been atrophy compared with later time points.

The activation maps provided insights into potential regions that the models use to differentiate atrophic optic nerve scans across the conditions studied, with potential clinical and research implications. These findings might offer clues into the pathological mechanisms underlying these conditions by identifying key anatomical areas for investigating disease-associated structural changes, though further studies are necessary to confirm whether these findings are generalizable across broader clinical populations.

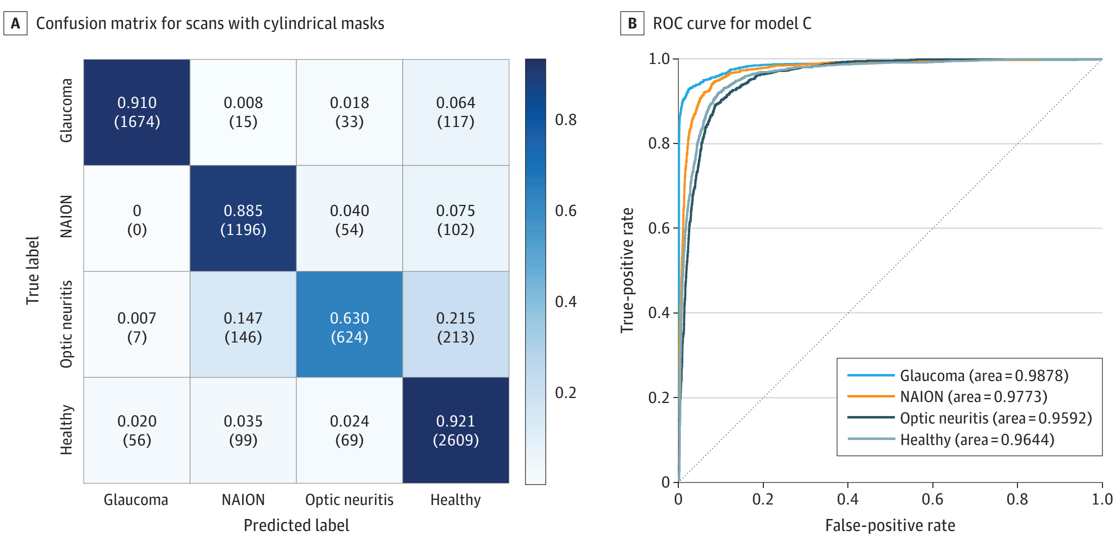
An advantage of a DL approach using the entire ONH volume is the model's ability to identify critical diagnostic changes and relationships across all B-scans comprising the volume of tissue sampled, potentially uncovering findings such as spatial patterns that may be missed when only a few 2D slices are analyzed. Unlike traditional methods, the model does not rely on selecting specific B-scans, even

Figure 3. Confusion Matrix and Receiver Operating Characteristic (ROC) Curve for Model B




A, Confusion matrix for scans with a cylindrical mask centered on and occluding the optic nerve head over all folds. Model B achieved an overall accuracy of 85.9% with a precision of 0.95, 0.81, 0.77, and 0.86, as well as a recall of 0.87, 0.86, 0.67, and 0.92, and F1 scores of 0.90, 0.83, 0.72, and 0.89 for eyes with glaucoma, nonarteritic anterior ischemic optic neuropathy, optic neuritis, and healthy eyes, respectively. B, The ROC curve for model B demonstrates high discriminatory capability across all classes. The macro-average area under the curve is 0.970.

Figure 4. Confusion Matrix and Receiver Operating Characteristic (ROC) Curve for Model C




A, Confusion matrix for scans with a cylindrical mask centered on and occluding the optic nerve head over all folds. Model C achieved an overall accuracy of 87.0% with a precision of 0.96, 0.82, 0.80, and 0.86, as well as a recall of 0.91, 0.89, 0.63, and 0.92, and F1 scores of 0.94, 0.85, 0.71, 0.89 for eyes with glaucoma, nonarteritic anterior ischemic optic neuropathy, optic neuritis, and healthy eyes, respectively. B, The ROC curve for model C demonstrates high discriminatory capability across all classes. The macro-average area under the curve is 0.972.

those with the highest utility, thereby aiding (or eliminating) the need for a clinician's expertise in choosing the most high-yield slices. Further, volume studies include the relationship between structures across the entire area, structures at multiple depths in the ONH and PPR, and shape information. OCT machines also offer greater standardization compared with fundus photo cameras, which can vary in resolution, color gradations, brightness, retinal pigmentation, contrast, and field of view. OCT devices consistently image the same region of the eye with uniform field of view and resolution without requiring pupil dilation, which is optimal for DL applications.

## Limitations

Our investigation had limitations. Although our RNFL thickness inclusion threshold of less than 100 µm ensured that we excluded residual, long-standing swelling in NAION and optic neuritis, this approach may have introduced bias against participants who had baseline thicker RNFLs. Conversely, it is possible that we included eyes with baseline thinner RNFLs that experienced acute, but mild, swelling attacks. However, these cases are uncommon. Given the retrospective nature of this study, baseline scans were unavailable for many eyes, making a 100-μm threshold a reasonable compromise between these considerations. Another notable factor is the difference with regard to sex between the disease groups and controls, particularly in the case of optic neuritis, where the majority of cases are in women. While the age of our participants was somewhat matched to the controls (particularly the University of Iowa for younger participants and VIP and Quark207 trial fellow eyes for older participants), there remained a lack of middle-aged controls and a more pronounced sex discrepancy as all University of Iowa young control participants were male. Including a broader range of young and middle-aged controls, particularly women, would improve generalizability and ensure a more balanced demographic representation, helping to minimize potential biases in the analysis. The same issues can be considered for race and refractive error as the optic disc size and appearance and RNFL vary with these factors. Quark207 trial fellow eyes had a smaller cup-disc ratio, potentially introducing a bias toward reduced ONH dimensions.[20] However, this systematic difference was not present in our other healthy cohorts.

Another limitation is that, given the complexity of the data, we have relatively few scans for each disorder, particularly for optic neuritis (990 scans), which limits the model's generalizability. It is possible that the variability in patterns of atrophy is greater than what our dataset captures, even with the use of a stratified 5-fold approach to maximize data utilization. Expanding the dataset would capture a broader spectrum of atrophic changes and improve the robustness of the model. Certain data sources in our study were geographically narrow; for instance, while glaucoma was one of our most abundant data sources, all data were sourced exclusively from the University of Iowa Health System in the Midwest United States. Our optic neuritis cases were sourced from a single neuro-ophthalmology clinic in New York City. While the high degree of accuracy observed for glaucoma may show the possibility of systematic differences based on location or patient population characteristics, the lower accuracy for optic neuritis and its significant overlap with healthy eyes suggest that this may not necessarily be the case for all optic nerve disorders. The data from healthy eyes, which come from diverse areas and demographics, indicate that factors beyond geographic or population-specific differences likely contribute to the model's challenges in distinguishing optic neuritis. However, other groups like NAION or healthy eyes were sourced from many sites in different countries. Further, model A still shows efficacy in discriminating between eyes with NAION and the healthy fellow eyes, with the scans of both acquired using the same device minutes apart.

We acknowledge that the disease groups were not matched for the degree of visual dysfunction. As such, the models could be biased based on the vision loss severity, though there is a substantial overlap among groups. As each disease affects the ONH differently, it would be impossible to get matching vision function sets without creating bias about the OCT structure data. For example, glaucoma with significant visual acuity impairment typically has a total cup, and optic neuritis with significant visual acuity loss has severe optic atrophy.

Future work could explore the integration of macular scans in addition to the ONH, allowing for DL to find the relationship between the types of atrophy for the 2 areas. Alternatively, an already segmented scan would provide additional information for the model. This would likely improve accuracy across all cases, as a segmented scan would offer more detailed data to help differentiate between conditions. To identify the exact spatial patterns of atrophy, methods such as variational autoencoders can explore the range of possible spatial patterns of atrophy across different datasets, while approaches like archetypal analysis or principal component analysis can uncover commonalities both within instances of a single disease and across multiple diseases.[21-27]

## Conclusions

Our findings highlight that optic nerve diseases exhibit distinct patterns of atrophy, which could support retrospective diagnostic efforts in cases lacking formal diagnoses. While many clinicians struggle to diagnose solely based on RNFL thickness patterns because of their subtlety and the overlap between conditions, this tool may assist in identifying disease-specific signatures. Furthermore, this method may support research into the progression of structural changes of these diseases, inform evolving diagnostic criteria, and offer potential utility in settings with limited access to subspecialty care.

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